

Production .957

Full Technical Architecture

A complete description of the detection ensemble, graph construction, motion and gap repair, division ownership, UniGRAFT, continuation repair, cleanup, validation, data packages, training provenance, and streaming implementation.

01

Detect

P1/P2 3D evidence

02

Construct

ILP graph

03

Repair

Motion and gaps

04

Divide

Geometry + UniGRAFT

05

Own

EdgeGRAFT +
CandidateGRAFT

06

Validate

Atomic output

Executive summary

What the system is designed to accomplish

0.957

HIDDEN PRODUCTION SCORE

11

ATTACHED ARTIFACT PACKAGES

2 + N

GPU PRODUCERS + CPU POOL

1 / 2

MAX IN-DEGREE / OUT-DEGREE

Core idea

The system does not depend on a single end-to-end network. It constructs a strong base graph from two 3D detection-and-association models, then applies specialized graph transactions. Each helper has a narrow role: repair motion ownership, bridge one-frame gaps, propose divisions, arbitrate source cardinality, repair continuation ownership, or preserve biologically plausible tracks through cleanup.

Production philosophy

Learned evidence and deterministic geometry are deliberately separated. Model outputs generate probabilities and candidate features; graph helpers commit only complete transactions that satisfy time, degree, distance, and ownership contracts. Every movie is processed end to end, every shard is validated before promotion, and the output is assembled only after dataset completeness is confirmed.

FIVE ARCHITECTURAL PRINCIPLES

- Evidence reuse: P1, P2, and Model C evidence is exported once and reused by later stages.
- Atomic ownership: division and continuation changes are committed as complete graph transactions, never as half-edits.
- Sparse-label safety: unannotated candidates are excluded from supervised negatives in the learned graph heads.
- Topology first: accepted graph structure is validated before coordinate smoothing and before CSV promotion.
- Operational determinism: hashes, manifests, stable candidate ordering, bounded caches, and atomic shard replacement make the run reproducible.

What the pipeline predicts

A graph in physical 3D space and time

Node

A retained cell center at one movie frame. Coordinates are stored as z, y, x in voxels and interpreted using 1.625, 0.40625, 0.40625 μm per voxel.

Continuation edge

A directed $t \rightarrow t+1$ identity claim. It says one cell persists into the next frame. A target may have at most one parent.

Division fork

Two outgoing $t \rightarrow t+1$ edges from one source. The two daughters are owned as one event; partial daughter installation is forbidden.

Graph invariants

- Every edge references two retained nodes.
- Every edge advances exactly one frame.
- Maximum in-degree is 1; maximum out-degree is 2.
- A two-child source is treated as a division transaction.
- Coordinates may be smoothed; topology may change only in named graph stages.
- The final graph must be non-empty and serialization-complete.

Why physical units matter

The raw volume is anisotropic: one z voxel spans four times the physical distance of one y or x voxel. All geometric gates therefore operate in micrometres. This prevents a one-voxel z change from being treated as equivalent to a one-voxel lateral change and motivates z-damped motion extrapolation.

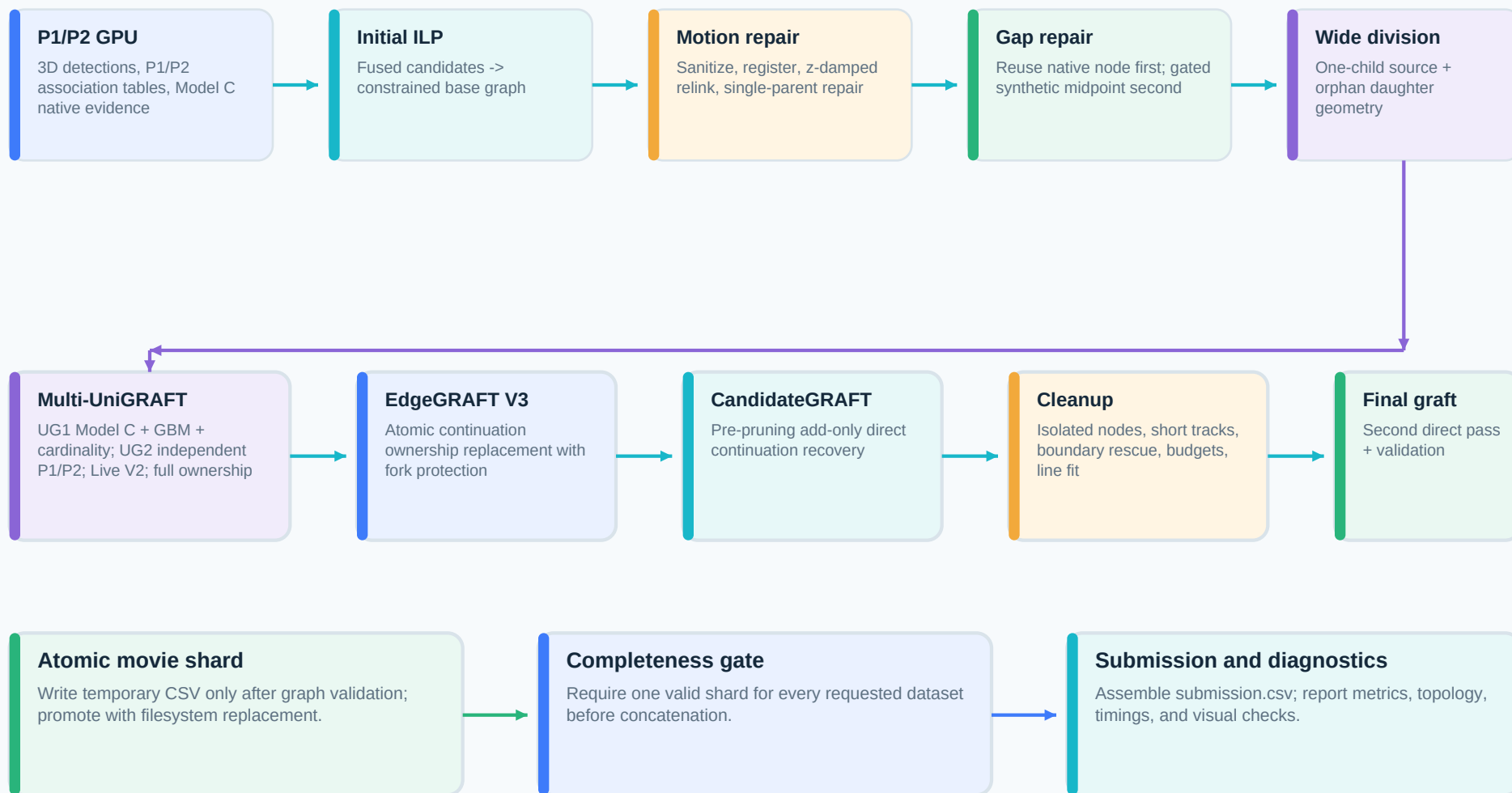
Scoring interface

The final report separates raw edge Jaccard, adjusted edge Jaccard, node recall, and division Jaccard. The competition score combines adjusted edge quality with 0.1 times division Jaccard. The node budget is therefore a graph-quality control, not a substitute for detecting and tracking real cells.

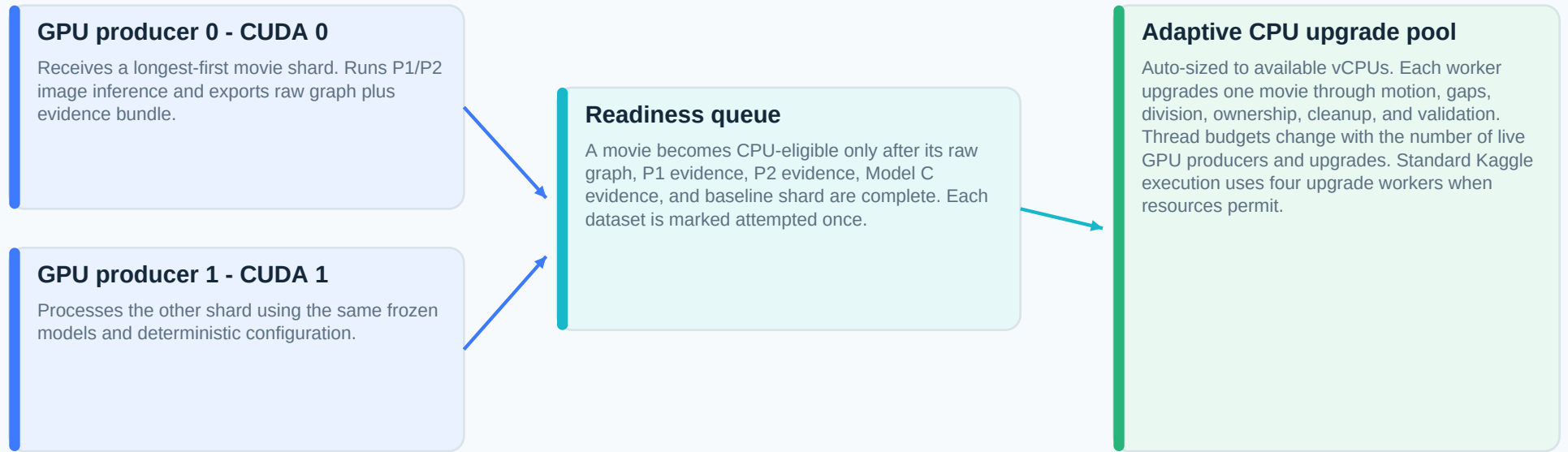
Updated production inference flow

Current .957 order - replaces the earlier worker diagram

Two GPU producers create raw evidence; CPU workers upgrade one movie at a time as soon as its bundle is ready.

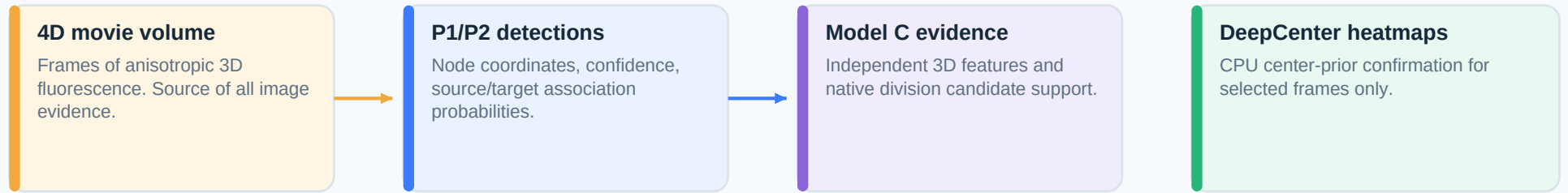


Streaming scheduler and worker lifecycle



NO DUPLICATE WORK ON THE NORMAL PATH

Work item	Frequency	Production behavior
Image inference	Once per movie	P1/P2 and evidence tables are created in the GPU producer.
Combined graph upgrade	Once per movie	All active helpers execute in the CPU worker.
DeepCenter	On-demand	Frame heatmaps are cached and shared by gap and division confirmation.
Evidence loading	Reused	Uncompressed archives avoid repeated decompression in downstream stages.
Baseline/emergency	Deadline only	Recovery paths run only when the normal combined upgrade cannot finish.



Evidence consumers

Evidence object	Where it is used
Detection logits	Candidate fusion, sub-voxel localization, initial node population
P1/P2 native associations	ILP, motion confidence, source cardinality, independent UG2, EdgeGRAFT, CandidateGRAFT
Model C native associations	Model C pair decoder, division GBM, source-cardinality features
Raw ILP graph snapshot	Motion relink substrate and EdgeGRAFT ownership comparison
Frame registration	Motion prediction and Motion V1 feature vector
DeepCenter heatmap	Long-gap confirmation and wider-geometry daughter confirmation
Current graph topology	All transaction eligibility, conflict checks, cleanup, and validation

P1/P2 detection and association ensemble

P1 - primary checkpoint

A 3D UNet produces volumetric features and a node-transformer association head scores temporal candidates. The deployed support package also supplies the inference repository. The checkpoint is hash-verified before execution. Its repackaged artifact does not retain a complete movie-split record, so this document does not claim an unseen P1 validation split.

P2 - independent seed

The same detector/association family trained with an independent seed. Its released training record covers the 199 training movies. It is used as complementary evidence, not as a replacement owner. Independence at initialization supplies useful probability disagreement and ranking diversity.

Why the ensemble is asymmetric

P1 remains the backbone. P2 contributes 0.475 to detection fusion, 0.15 to continuation evidence, and only in the low-margin regime. This preserves the strong P1 graph while allowing P2 to resolve uncertain detections and associations.

EXACT INFERENCE CONTRACT

Parameter	Value	Function
Detection threshold	0.96875	Peak acceptance after scale-aligned logit fusion
Detection fusion	P2 weight 0.475	P1 remains primary
Candidate retention	>= 90% of P1 population	Prevents the second seed from shrinking the base population
P2 edge contribution	0.15	Applied through low-margin consensus
Low-margin ceiling	0.35	Restricts P2 ownership influence to ambiguous P1 cases
Edge candidate threshold	0.48	Candidate generation threshold
Bidirectional support	0.30	Harmonic fusion of forward and reverse probabilities
Sub-voxel refinement	+/- 0.49 voxel	Parabolic peak adjustment without changing network sampling lattice

Initial ILP graph construction

Turning local probabilities into a legal base graph

Input

Fused node candidates, P1/P2 continuation scores, appearance/disappearance alternatives, and the base division option. The ILP is the first global ownership decision: candidate rows become one coherent movie graph rather than independent edge classifications.



Output

A directed graph whose selected edges obey the base optimizer constraints. The raw result is preserved as evidence because later stages may compare, replace, or protect its continuation ownership.

ILP term	Value	Role
Edge weight	-1.0	Rewards supported temporal association
Appearance	0.0	No extra appearance charge in the optimizer
Disappearance	1.5	Cost for ending a track
Division	1.0	Base cost assigned to a fork choice

Important ownership boundary

The ILP is not the final graph. The registered motion stage is allowed to replace its continuation edge set, while the later division and ownership helpers operate on the resulting graph. For that reason, the pipeline stores both the evolving graph and the original evidence needed for downstream arbitration.

Registered z-damped motion relinking

Whole-frame assignment with a learned residual



Motion V1 training provenance

Small MLP: 23 -> 64 -> 32 -> 1 with SiLU activations and 0.05 dropout. It learned a bounded residual in [-2,+2] on candidate continuations derived from 199 historical A+B proposal files. Split: 190 training movies and 9 validation movies. Candidate cache: 354,255 training rows and 540,613 validation rows. AdamW, learning rate 0.002, batch 8,192, focal-weighted BCE, up to 40 epochs with patience 7. Unknown graph rows were not converted into division behavior because this model only scores continuations.

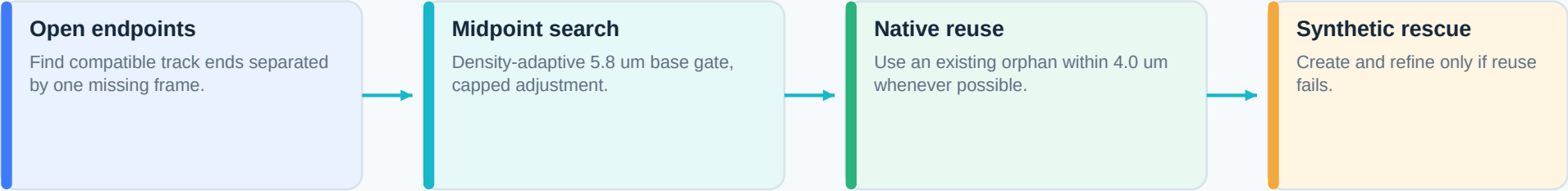
Current deployment rule

The current P1/P2 graph differs from the historical proposal population, but the checkpoint remains the hidden-validated production component. The runtime preserves its one-step velocity input while the geometric predictor uses a three-step history. Assignment cost = registered motion distance + 0.05 raw distance - native edge support - learned residual. A 7.5 geometry-only refusal cap prevents confidence from purchasing an implausibly long jump.

Control	Value	Reason
Velocity axes	[0.0, 0.45, 0.47]	Suppress noisy z extrapolation; preserve lateral motion
History	3 steps	Mean trajectory for geometric prediction
Corrector history	1 step	Preserves trained 23-feature contract
Tight preference	3.0	Joint assignment favors strong local candidates
Dense-frame ceiling	4,000 nodes	Above this, keep raw ILP edges for tractable assignment

One-frame gap recovery and DeepCenter

Reuse a real node before synthesizing one



DeepCenter model

DeepCenterUNet3D is an auxiliary 3D center-heatmap network, not a tracker. Sparse cell centroids were rendered as Gaussian targets. Training used positive-unlabeled weighted BCE: dark background carried normal negative weight, bright unlabeled voxels carried low weight, and labeled centers carried 12x positive weight. Volumes were XY pooled by 4 and normalized per frame. The deployed best checkpoint is epoch 2 by validation loss; later training states were not substituted.

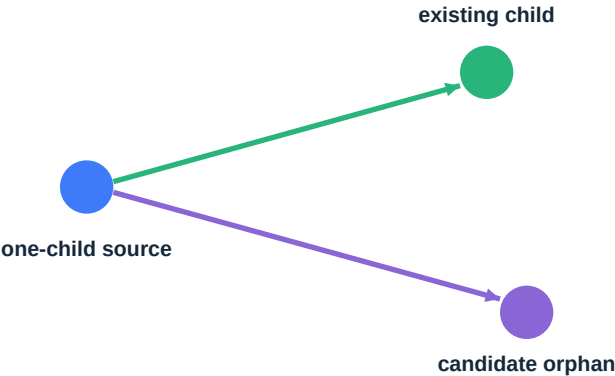
How production uses it

DeepCenter never seeds a free population. It scores only targeted locations. A long synthetic gap with total span ≥ 8.5 um must reach 0.25 in a local heatmap window. The same bounded frame cache is reused by the later division confirmation stage, avoiding a second network evaluation for the same frame. This turns a broad detector into a narrow evidence check.

Control	Value	Purpose
Synthetic cap	min(3% of movie, 2,000 nodes)	Bounds node growth
Local refinement	z +/-1, y/x +/-3; max shift 3.2 um	Moves midpoint toward local signal
Density context	3 neighbors; reference 6.5 um	Small local adjustment only
Heatmap cache	8 frames	Shared with wide-division confirmation
Device	CPU	Keeps both GPUs available for P1/P2 production

Wider-geometry division pass

Deterministic daughter recovery before UniGRAFT



Transaction question

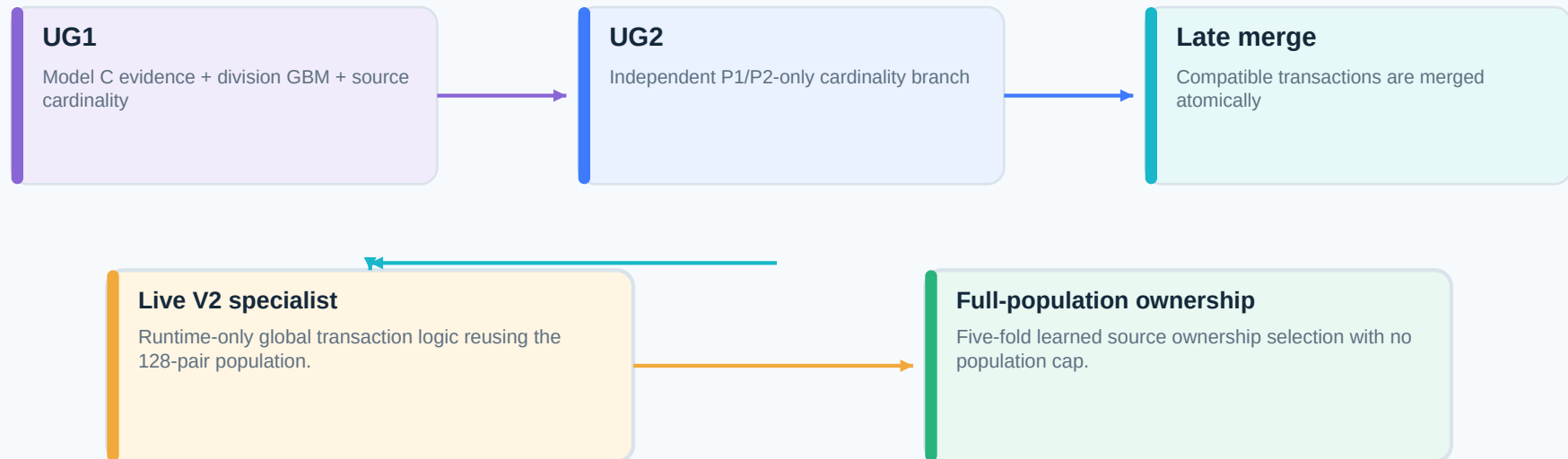
Does one already-linked parent have geometric and appearance support for a second daughter? The stage proposes one complete two-child event. It does not train a classifier and does not bypass later ownership arbitration.

Gate	Frozen value	Biological/graph purpose
Parent to candidate daughter	$\leq 9.0\text{ }\mu\text{m}$	Local daughter reach
Daughter pair distance	$\leq 14.0\text{ }\mu\text{m}$	Sister compatibility
Existing child distance	$\leq 10.0\text{ }\mu\text{m}$	Protect plausible first daughter
Symmetry	0.60	Rejects strongly imbalanced daughter geometry
Orphan ownership	Mutual nearest neighbor	Prevents opportunistic daughter stealing
t+2 divergence	$\geq 2.25\text{ }\mu\text{m}$	Requires daughter trajectories to separate
DeepCenter	≥ 0.25	Confirms the candidate center
Proposal caps	0.0076/frame; 0.00375/global	Bounds event density

What UniGRAFT is

A nested division and ownership system, not a detector

UniGRAFT converts multiple sources of division evidence into legal graph transactions. Its unit of decision is a source and an option set: CONTINUE, or DIVIDE(daughter A, daughter B). It evaluates source occurrence, daughter identity, existing ownership, and topology together. The term Multi-UniGRAFT refers to the ordered combination of independent branches plus a final ownership resolver.



ATOMIC TRANSACTION CONTRACT

- A division is installed as two daughter edges or not installed at all.
- No target may receive two parents; no source may exceed two children.
- One event is retained per linked source tube where required.
- Existing accepted forks are protected unless an authorized ownership transaction supersedes them.
- Stable candidate ordering and a 128-pair cap keep training and serving aligned.
- The preceding wider-geometry pass is visible to UniGRAFT and may be kept, rejected, or superseded.

Model C evidence model

Independent 3D UNet/node-association checkpoint used only to describe division candidates. Detection threshold 0.99, four-view flip TTA, 3.0 um pooling, 20.0 um evidence radius, and up to 16 targets per source. Native Model C nodes map to the base graph within 6.0 um.

Division GBM cascade

Pair model scores 78-feature daughter pairs. Per-source aggregation appends best-pair statistics to 41 source features, producing a 123-feature source decision. Models are histogram gradient boosting with grouped five-fold pair-score parity.

Source-cardinality head

Jointly normalizes one CONTINUE option against retained DIVIDE pairs. Input: 41 source features and 150 pair features: 78 geometry + 24 Model C + 24 P1 + 24 P2. Threshold 0.40, maximum 128 pairs, stable geometry order.

TRAINING RECORD

Learned component	Split	Population	Sparse-label policy
Division GBM	199 movies	1,095,856 pairs / 479 positive; 127,398 sources / 436 positive	Unknown rows excluded; grouped five-fold pair probabilities
Model C combined decoder	175 train / 20 held	102 pair features / 144 source features	Threshold 0.96; four practice clips not used for fitting
Source cardinality	5 grouped folds	2,179 parameters; 130 positive sources; 92,675 confirmed continuations in full fit	60 epochs; grouped source-local softmax; no NULL class

Why UG1 exists

P1/P2 contain substantial daughter identity evidence, but ordinary edge thresholds cannot decide whether a source should continue or divide. UG1 adds an appearance-rich third evidence family, then makes the source-level cardinality decision jointly so a moderately strong continuation does not automatically suppress a biologically supported fork.

UG2 - independent P1/P2 branch

A separately trained source-cardinality head that removes Model C from the input. It uses the same exact-frame source population, 41 source features, 126 pair features, five grouped movie folds, a 128-pair cap, and a frozen 0.51 threshold. Its independence allows compatible events missed by UG1 to enter the late merge.

Live V2 - runtime specialist

Contains no model weights, annotations, dataset/source IDs, or cached decisions. It reuses the current 128-pair V2 population and existing evidence. It applies deterministic global transactions after UG1/UG2. Validation on 42 changed movies recorded +8 TP, 0 FP, and -8 FN on that matched subset.

Full-population ownership

A five-fold ExtraTrees source owner with 14 competition and geometry features. It considers every eligible source, has no population cap, selects the best candidate per source above 0.947082, and never treats unannotated sources as negative training rows.

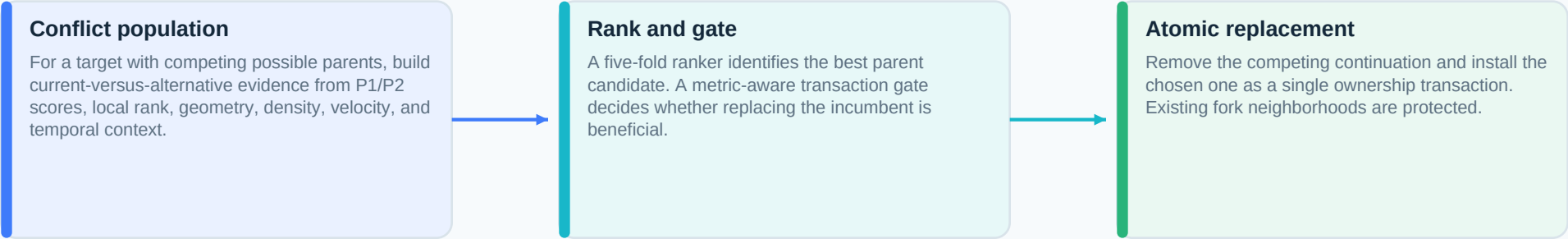
Component	Supervision	Recorded evidence	Frozen form
UG2 head	Exact division time + confirmed continuations	6 / 4 / 11, J=0.2857 on held panel	Frozen threshold 0.51
Live V2	No learned fit	+8 TP / 0 FP / -8 FN on 42 changed movies	Runtime transaction bundle
Ownership model	Known-only rows, 5 grouped folds	2,089,875 candidates; 654 OOF-selected sources	400 trees, depth 3, leaf >=20

Why the branches remain separate

Combining all evidence into one monolithic model would make disagreement invisible. The nested design preserves complementarity: UG1 can use Model C morphology, UG2 can act when native P1/P2 rankings are clearer, Live V2 can recover a structurally recognizable event without another image pass, and the ownership model resolves the broad residual population under one graph contract.

EdgeGRAFT V3 continuation ownership repair

Replace the wrong parent, not the entire graph



TRAINING AND SCALE



Layer	Mechanism	Production contract
Parent ranker	HistGradientBoosting classifier	Five grouped folds; 86,864 OOF targets; accuracy 0.9652 -> 0.9818
Transaction gate	Metric-positive vs metric-negative replacements	97,484 transactions; 3,883 non-neutral labels; neutral rows excluded
Feature families	P1/P2 probability and rank; raw graph; density; motion; incoming/outgoing context	New, old, and delta features compare the replacement directly
Fork policy	Protect source, daughters, parent, and immediate daughter successors	Continuation repair cannot silently dismantle division topology

CandidateGRAFT and retention guards

Add direct continuations without taking ownership away

CandidateGRAFT direct model

A lightweight classifier scores one-frame P1/P2 candidates using eight features: P1 probability, P2 probability, max, mean, model presence, winner count, best rank, and distance. It may add an edge only when the source has no child and the target has no parent. It never deletes an edge and never creates a fork.

Training record

Five grouped movie folds over 29,231 population rows. The fit used 348 annotated known rows: 304 positive and 44 negative. Unannotated rows were not negatives. Average precision was 0.939; the OOF-frozen gate is 0.90. Final refit preserved the same threshold and eight-feature contract.

Why it runs twice

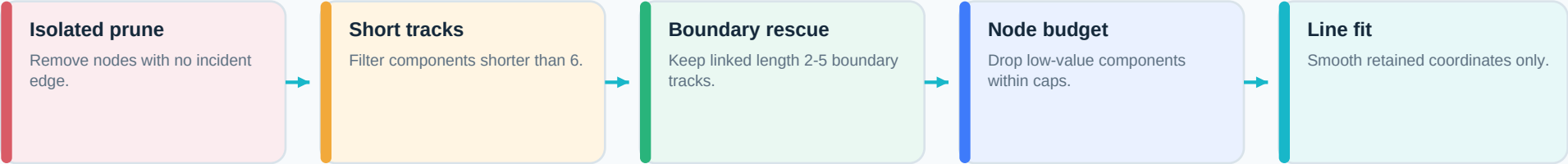
Operation	Placement	Purpose
Pre-pruning pass	After EdgeGRAFT	Reconnects legitimate fragments before isolated-node and short-track removal can erase them
Final pass	After cleanup and line fit	Uses the surviving node population to recover direct slots exposed by cleanup
Retention guard	After each ownership repair	Restores named transitions if a helper attempted to change protected P2-zero ownership

Degree-slot logic

CandidateGRAFT is deliberately different from EdgeGRAFT. EdgeGRAFT may replace an incumbent parent after a learned conflict decision; CandidateGRAFT may only fill a completely open source-to-target slot. This separation prevents a broad continuation classifier from competing with installed divisions or from undoing high-confidence ownership.

Cleanup, boundary rescue, node control, and smoothing

Destructive operations occur only after graph rescue



Helper	Frozen rule	What it does
Isolated-node pruning	Degree zero	Removes detections that never participate in a track
Short-track filter	Minimum component length 6	Suppresses unstable fragments; any component containing a fork is protected
Boundary rescue	Length 2-5 at movie boundary	Retains a track that cannot reach length 6 because observation starts or ends
Normal node budget	1.00 x detected; max drop 3%	Limits post-processing growth while preserving fork components
Dense node budget	>=40,000 detections; 0.98 ratio; max drop 4%	Tighter control for very dense movies
Line-fit smoothing	Window 3; weight 0.8	Reduces coordinate jitter without changing edges

Ordering rationale

CandidateGRAFT runs before this block so a true direct continuation can rescue a fragment. Division-containing components are protected from short-track and budget removal. Line fitting is last because it changes coordinates but not topology; the final CandidateGRAFT pass then checks only the retained nodes.

Validation, atomic shards, and final output

A graph is promoted only after it is complete

Graph contract

- All edge endpoints exist.
- Every edge connects consecutive frames.
- In-degree ≤ 1 and out-degree ≤ 2 .
- Protected daughter pairs and ownership contracts remain intact.
- Wider-geometry division retention is counted after each major stage.
- The movie graph is non-empty.

Output contract

- Write a temporary per-movie CSV shard.
- Serialize only after the full graph contract passes.
- Replace the existing shard atomically.
- Require one shard for every test dataset.
- Concatenate in deterministic dataset order.
- Write metrics, run statistics, topology counts, and visual diagnostics.

FAILURE CONTAINMENT

Condition	Action	Safety property
Dependency failure	Stop before inference	Missing helper or wrong checkpoint cannot hide behind a fallback
Worker exception	Do not promote partial graph	Temporary shard remains uncommitted
Deadline pressure	Use validated recovery shard	Normal combined result is preferred; baseline/emergency is last resort
Missing dataset shard	Block assembly	Submission is not silently made incomplete
Topology violation	Reject movie result	Bad ownership cannot enter submission.csv

Attached dataset packages - neural and division evidence

Logical production names and responsibilities

Attached package	What it contains	What production uses it for
P1 inference support package	3D UNet + node transformer checkpoint, inference repository, weight bundle	Primary detections, association logits, raw graph evidence
P2 independent temporal package	Independent seed checkpoint for the same detector/association family	Secondary detection fusion, low-margin edge consensus, diversity features
Model C combined division package	Model C checkpoint, pair/source GBMs, safe runtime, evidence exporter	Morphology-rich native division candidates and UG1 source decisions
Source-cardinality V2 package	Grouped option head and graph transaction runtime	CONTINUE versus DIVIDE normalization per source
Independent P1/P2 UniGRAFT package	P1/P2-only cardinality checkpoint	UG2 division branch independent of Model C
DeepCenter center-prior package	DeepCenterUNet3D checkpoint, config, coordinate contract	Targeted confirmation of long gap nodes and wide daughters

Package resolution and integrity

The notebook resolves each attached package by declared artifact identity, verifies required files and checkpoint hashes, checks DeepCenter epoch metadata, materializes the inference repository, and imports helper classes before any producer or worker starts. This is a dependency gate, not a runtime fallback.

Attached dataset packages - graph ownership and runtime

every package has one bounded production role

Attached package	What it contains	What production uses it for
Live V2 global bundle	Runtime module only; no model weights, labels, dataset IDs, or cached decisions	Post-UG1/UG2 missed-division transactions
Full-population ownership package	Five ExtraTrees fold models, deploy spec, ownership runtime	Broad source ownership selection without an inference population cap
EdgeGRAFT V3 package	Five ranker folds, five metric-gate folds, component/runtime modules	Atomic continuation parent replacement after division ownership
CandidateGRAFT direct package	Eight-feature classifier, frozen 0.90 gate, add-only runtime	Fill unowned one-frame continuation slots before and after cleanup
Motion Corrector V1 package	23-feature MLP checkpoint with normalization and residual scale	Learned correction to registered motion assignment cost

COMPETITION INPUT

Movie volumes and metadata

The competition input supplies the test movie volumes and required dataset identifiers. Frames are read by the P1/P2 producer and selectively by the CPU DeepCenter helper. All graph coordinates retain the competition z, y, x voxel convention; all geometric decisions convert to physical units before gating.

Training provenance and inference status

What was learned, what was fitted, and what is deterministic

Component	Learned object	Training/selection record	Production use
P1	3D detector + association model	Checkpoint package; exact split not retained in repackaged record	Frozen neural inference
P2	Independent detector + association seed	Available release record lists 199 training movies	Frozen neural inference
Model C	3D division evidence model	Independent checkpoint; 4-view TTA at serving	Evidence only; does not own base graph
Division GBM	Pair and source classifiers	199 movies; grouped 5-fold pair probability construction	Frozen pair -> source cascade
Source cardinality	Low-capacity grouped option head	5 grouped movie folds; exact-time labels; unknown sources excluded	Frozen 0.40 source transaction gate
UG2 cardinality	Independent P1/P2 option head	Same exact-frame 175/20 population; 5 grouped folds	Frozen 0.51 gate
Motion V1	23-feature residual MLP	199 proposal movies; 190/9 split; historical A+B candidate population	Frozen checkpoint on current motion cost
DeepCenter	3D center-prior UNet	Sparse Gaussian centers; positive-unlabeled weighted BCE; 10% validation	Targeted CPU confirmation only
Ownership	ExtraTrees source owner	Known-only rows; 5 grouped folds	Hash-routed unseen fold model
EdgeGRAFT	Boosted ranker + transaction gate	175 movies; grouped folds; neutral transactions excluded	Atomic continuation replacement
CandidateGRAFT	Eight-feature classifier	348 known fit rows; 5 grouped folds; unknown excluded	Add-only continuation
Wide division / Live V2 / cleanup	Deterministic graph logic	No fitted weights	Frozen rules and topology contracts

Frozen production constants

Values read from the live .957 notebook

Control	Value
Voxel scale z/y/x	1.625 / 0.40625 / 0.40625 um
Detection threshold	0.96875
P2 detection / edge weights	0.475 / 0.15
Low-margin / edge threshold	0.35 / 0.48
Candidate retention / harmonic	0.90 / 0.30
ILP edge / appearance	-1.0 / 0.0
ILP disappearance / division	1.5 / 1.0
Edge maximum	14.0 um
Motion tight / relaxed	6.0 / 10.0 um
Velocity axes z/y/x	0.0 / 0.45 / 0.47
Motion refusal	7.5 geometry-only

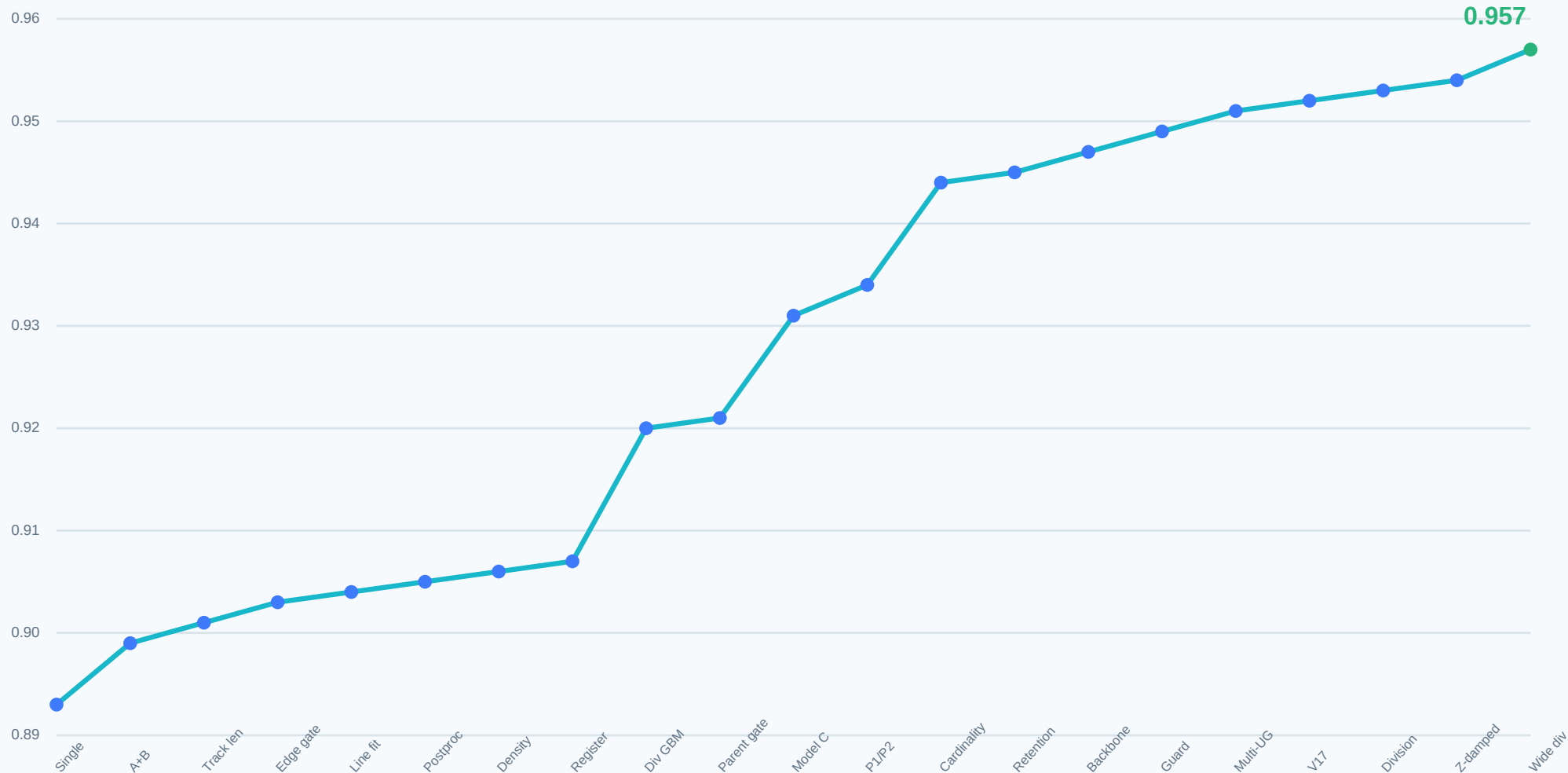
Control	Value
Gap close / native reuse	5.8 / 4.0 um
Synthetic gap cap	3% or 2,000
DeepCenter threshold	0.25
DeepCenter long-gap span	8.5 um
Wide division parent/daughter	9.0 um
Wide division sister	14.0 um
Wide division divergence	2.25 um
Minimum track length	6
Boundary rescue length	2-5
Normal / dense node ratio	1.00 / 0.98
Line fit weight / window	0.8 / 3

Configuration discipline

These values are centralized in one configuration cell and exported to child processes. Speed controls are separated from graph controls. Runtime acceleration is installed only after bit-for-bit parity tests; mixed precision remains disabled because it changes detector logits and therefore the node population.

Production score progression

Milestones retained in the current architecture



Interpretation

The score progression is a lineage, not a claim that every isolated delta is additive. The current .957 system retains the successful concepts shown here and adds the wider-geometry division pass immediately before Multi-UniGRAFT.

What to monitor in a production run

Counters that prove the intended path executed

Run area	Required observation	Meaning
Preflight	All artifact roots resolved; all hashes and expected files match	Any missing helper is fatal before inference
GPU production	One raw graph and three evidence tables per movie	No dataset dispatched before its complete bundle exists
CPU upgrades	Combined upgrades == total datasets	Confirms normal production path for every movie
Wide division	Proposed, retained after UniGRAFT, retained final	Shows where new daughter transactions survive or are superseded
EdgeGRAFT	Candidate decisions, applied ownership replacements, guarded reverts	Distinguishes model action from guard enforcement
CandidateGRAFT	Eligible rows, selected additions, conflicts skipped; pre and final separately	Proves add-only continuation recovery executed
Cleanup	Isolated nodes, short components, boundary rescues, budget drops	Makes destructive graph changes auditable
Topology	Nodes, edges, forks, degree violations, nonconsecutive edges	Must finish with zero contract violations
Performance	Per-stage seconds and worker wall time	Identifies tail movies without changing inference

Operational acceptance condition

A successful notebook finish is not sufficient by itself. Promotion requires the complete dependency check, one validated shard per dataset, all combined upgrades accounted for, zero final graph-contract failures, and a submission assembled from the promoted shards.

System summary

Production .957 is a complete 4D cell-tracking architecture. P1/P2 generate the node and association evidence; the ILP creates the initial legal graph; motion and gap helpers repair continuity; the wider-geometry pass introduces additional division hypotheses; Multi-UniGRAFT resolves division occurrence, daughter identity, and source ownership; EdgeGRAFT repairs competing continuation parents; CandidateGRAFT fills open direct slots; cleanup controls fragments and node growth; validation and atomic shard promotion protect the submission. The design is modular, but the production order is part of the model contract.

Term	Meaning in this architecture
Candidate	A possible node, edge, or two-daughter transaction before graph commitment.
Ownership	The exclusive parent-child assignment represented by graph edges.
Cardinality	Whether a source ends, continues with one child, or divides into two daughters.
Transaction	A complete, conflict-checked graph mutation committed atomically.
Tube	A linked lineage neighborhood used to suppress duplicate division events.
Native evidence	Probabilities and features emitted by the checkpoint that produced the candidate population.
OOF	Out-of-fold prediction where a movie is scored by a model not fitted on that movie fold.
Sparse-label safety	Unknown or unannotated candidates are not automatically labeled as biological negatives.
GRAFT	A bounded graph-edit helper that adds or replaces ownership under an explicit contract.
UniGRAFT	The nested division/cardinality system that merges independent branches and resolves legal forks.

Basis: live production notebook, deployment manifests, model training summaries, and the production experiment registry.